

UST-OLOY, Russian Federation

WMO: 253250

Lat: 66.550N

Lon: 159.417E

Elev: 127

StdP: 99.81

Time Zone: 12.00 (E12)

Period: 90-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.7	-47.2	-51.0	0.0	-48.5	-49.6	0.0	-47.2	7.4	-24.6	5.9	-24.2	1.2	230	0.810

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	26.3	16.5	23.9	15.0	21.6	13.9	17.3	24.0	15.9	22.5	14.5	20.4	2.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	10.5	20.2	13.0	9.5	17.9	11.6	8.6	16.7	49.3	23.9	45.1	22.6	41.2	20.5	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	5.9	4.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.7	-36.5	-35.2	-25.4	-12.9	2.4	11.7	13.8	10.0	3.0	-11.8	-25.8	-35.5
	DBStd	19.77	7.46	6.50	7.93	7.20	5.24	3.92	4.32	3.93	4.39	7.42	8.10	7.75
	HDD10.0	8194	1442	1266	1098	687	240	27	14	50	212	675	1074	1409
	HDD18.3	10981	1701	1499	1356	937	494	199	151	258	460	933	1324	1668
	CDD10.0	268	0	0	0	0	4	79	132	50	2	0	0	0
	CDD18.3	12	0	0	0	0	0	2	11	0	0	0	0	0
	CDH23.3	263	0	0	0	0	0	51	192	21	0	0	0	0
	CDH26.7	56	0	0	0	0	0	7	46	3	0	0	0	0
Wind	WSAvg	1.6	0.9	0.9	1.4	1.6	2.3	2.1	1.9	2.0	2.0	1.5	1.0	1.1
Precipitation	PrecAvg	259	16	12	9	8	12	30	48	42	25	21	20	16
	PrecMax	366	57	37	28	33	51	107	109	104	86	50	51	43
	PrecMin	167	0	1	0	0	0	0	5	5	2	6	6	0
	PrecStd	51	11	8	7	7	9	21	27	23	19	9	11	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.7	-9.9	-0.2	6.1	20.4	27.4	30.1	25.7	17.4	4.2	-0.9	-7.2
		MCWB	-9.6	-10.6	-3.3	1.6	10.4	15.4	18.1	16.2	9.9	2.4	-3.1	-7.4
	2%	DB	-18.3	-18.3	-4.6	3.8	16.7	24.6	27.8	22.7	14.8	1.5	-7.9	-14.9
		MCWB	-18.9	-18.9	-6.6	0.0	8.4	14.0	17.2	15.2	9.1	-0.1	-8.5	-15.3
	5%	DB	-22.0	-21.5	-8.3	1.5	14.3	22.6	25.5	20.1	12.4	-0.2	-11.8	-19.1
		MCWB	-22.3	-21.9	-9.7	-1.5	7.0	13.1	16.6	13.8	7.8	-1.3	-12.2	-19.4
	10%	DB	-25.5	-24.1	-12.6	-0.9	11.5	20.5	22.9	17.7	9.8	-1.9	-15.0	-22.6
		MCWB	-25.7	-24.4	-13.6	-3.3	5.6	12.0	15.1	12.6	6.9	-3.2	-15.4	-22.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.4	-10.8	-2.9	2.0	10.7	16.5	20.3	18.5	11.5	2.5	-2.9	-7.2
		MCDB	-8.4	-9.9	-0.8	5.6	19.2	24.1	27.8	23.2	15.7	4.0	-1.1	-6.9
	2%	WB	-18.5	-18.9	-6.6	0.2	8.8	14.7	18.2	15.7	9.7	0.3	-8.4	-15.4
		MCDB	-18.3	-18.4	-4.7	3.4	16.0	22.4	25.6	21.4	13.3	1.1	-7.8	-15.0
	5%	WB	-22.2	-21.9	-9.7	-1.4	7.1	13.6	16.9	14.1	8.4	-1.4	-12.2	-19.5
		MCDB	-21.9	-21.6	-8.5	1.4	13.7	21.3	23.8	19.4	11.5	-0.1	-11.8	-19.2
	10%	WB	-25.7	-24.4	-13.5	-3.3	5.8	12.5	15.7	12.9	7.1	-3.3	-15.4	-22.8
		MCDB	-25.5	-24.1	-12.6	-0.8	11.3	19.6	22.1	16.9	9.6	-2.0	-15.0	-22.5
Mean Daily Temperature Range	5% DB	MDBR	6.7	8.7	12.9	13.8	10.2	11.3	11.3	10.2	7.6	7.6	7.4	5.6
		MCDBR	9.6	10.0	15.7	11.8	13.9	16.4	17.0	16.9	13.6	8.1	11.0	8.3
		MCWBR	9.4	9.6	13.9	8.4	7.6	7.1	7.9	9.3	8.0	6.9	10.4	8.1
	5% WB	MCDBR	9.6	10.0	15.4	11.0	13.0	14.8	15.5	14.6	11.6	7.8	11.0	8.5
		MCWBR	9.4	9.7	13.6	8.0	7.4	7.0	8.1	8.5	7.3	6.8	10.4	8.2
Clear-Sky Solar Irradiance	taub	0.167	0.234	0.299	0.354	0.392	0.375	0.449	0.422	0.326	0.270	0.186	0.174	
	taud	1.917	2.089	2.048	1.945	2.097	2.357	2.128	2.160	2.400	2.159	1.916	1.907	
	Ebn at Noon	254	622	735	780	795	831	738	714	730	551	192	20	
	Edh at Noon	15	52	95	138	132	105	129	113	69	48	14	3	
All-Sky Solar Radiation	RadAvg	0.08	0.74	2.55	4.98	5.73	5.58	4.90	3.69	2.00	0.79	0.16	0.01	
	RadStd	0.01	0.07	0.08	0.16	0.46	0.42	0.46	0.34	0.27	0.11	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page